TEXORA · LECTURE SHEET

Colour Fastness, Damage Estimation and Determination of Size

Colour is the first quality a buyer sees and the first to be complained about. This sheet covers how fastness to light, washing and rubbing is tested and graded with grey scales and the blue-wool scale, how chemical and physical damage in cellulosic fabrics is estimated, and how warp size is identified, removed and quantified.

TE-3201 · QUALITY EVALUATION OF FABRIC AND APPAREL · TOPIC 7 · LEVEL 3 TERM 2

1. What colour fastness means

Colour fastness is the resistance of a coloured textile to change — fading, deepening, hue shift — and to transferring colour to other materials (staining) when exposed to an agency: light, washing, rubbing, perspiration, water or heat. Every fastness test therefore produces *two* judgements: the **change in colour** of the specimen itself and the **staining** of an adjacent undyed material placed in contact with it. Both are graded, and a specification normally sets a minimum for each. Testing is done on conditioned fabric (65 % RH, 20 °C) by trained assessors under standard illumination, increasingly checked by instrumental colour measurement, but the visual grey-scale grade remains the contractual language of the trade.

2. The grey scales — ISO 105-A02 and ISO 105-A03

The grey scales convert a visual difference into a number from 5 to 1. Each scale is a set of paired patches whose contrast increases in steps.

- **Grey scale for colour change (ISO 105-A02)**: pairs of grey patches. One patch is always the original shade; the second patch in each pair differs by a standard contrast step. The assessor finds the pair whose contrast matches the contrast between the tested specimen and an untested piece of the same fabric. Grade 5 = no contrast (no change); grade 1 = very large contrast (severe change).
- **Grey scale for staining (ISO 105-A03)**: pairs of white patches, one always clean white, the other stained to a standard degree. The staining on the adjacent fabric is matched against this scale. Grade 5 = no staining; grade 1 = severe staining.

Half-grades such as 4–5 or 3–4 are used when the contrast falls between steps. The commercial interpretation is unforgiving: grade 4 is usually the minimum acceptable for apparel colour change, 3–4

to 4 for staining, and grade 3 or below on a main shade is a claim. Assessors work in a standard light cabinet, rest their eyes between judgements and grade independently — two assessors who differ by more than half a grade re-grade together. Instrumental grading with a spectrophotometer calculates the same grades from colour-difference values and is preferred for pass/fail disputes.

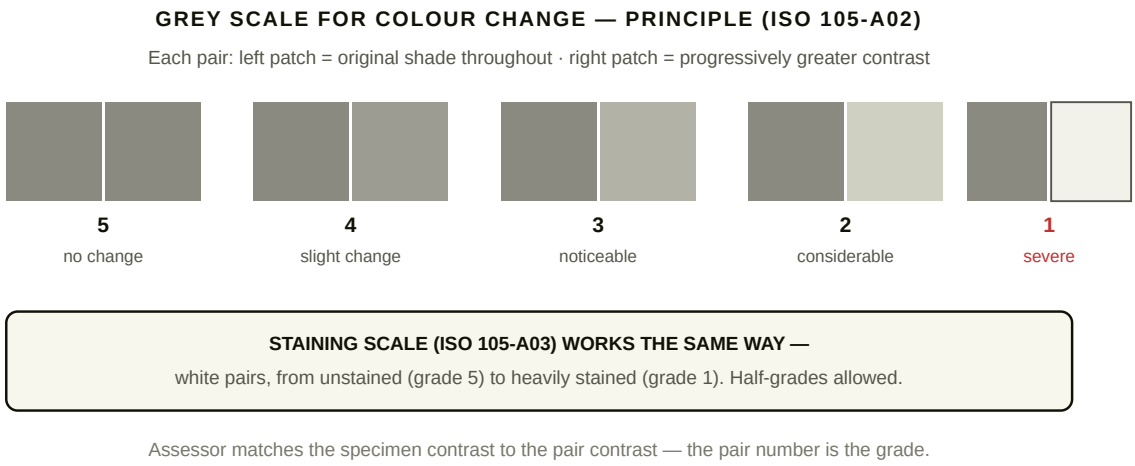


Figure 1 — Grey scale principle. Contrast between the two patches of a pair grows from grade 5 (none) to grade 1 (severe); the staining scale uses white pairs on the same logic.

3. Fastness to light — ISO 105-B02

Light fastness is tested against the **blue-wool scale**, a separate scale of 1 to 8. Eight standard wool fabrics dyed with blue dyes of progressively greater light fastness are exposed alongside the specimen — in ISO 105-B02 under a xenon-arc lamp that simulates daylight under controlled temperature and humidity, or in daylight methods under glass. Exposure continues until the specimen shows a defined contrast (grade 4, then grade 3 on the grey scale for colour change). The number of the blue-wool reference that shows the *same* contrast at the same time is the light-fastness grade: 1 = very poor (fugitive), 8 = excellent. Each step is approximately double the exposure life of the previous one, so grade 6 lasts roughly twice as long as grade 5.

Typical expectations: curtains and furnishings exposed daily need grade 6 or better; apparel commonly specifies 4–5; cheap fashion shades at grade 3 will visibly fade in one season of window display. Light fastness is primarily a property of the dye–fibre system and depth of shade — pale shades grade lower than deep shades of the same dye because the same absolute dye loss is a larger fraction of the colour present. Humidity and temperature during exposure matter: the test controls both, and a result quoted without its method (xenon-arc vs daylight) is incomplete.

4. Fastness to washing — ISO 105-C06

In ISO 105-C06 the specimen is sewn in contact with a **multifibre adjacent fabric** — a single strip containing bands of different fibres (typically acetate, cotton, nylon, polyester, acrylic and wool) — and washed in a laboratory machine with steel balls for mechanical action, in standard soap or detergent solution at a stated temperature, time and liquor ratio. The C06 series offers several severities (combinations such as 40 °C, 50 °C and 60 °C washes) chosen to represent one or several domestic washes. After rinsing and drying, the colour change of the specimen is graded with the A02 scale and the staining of *each band* of the multifibre strip with the A03 scale — the worst band is the one reported. The multifibre strip is an elegant device: one wash reveals at once whether loose dye will stain cotton, nylon or wool in a mixed wash load. A red garment that grades change 4–5 but stains the nylon band at 2 will ruin white nylon underwear in the same wash, and the report must show both facts.

5. Fastness to rubbing — ISO 105-X12

Rubbing (crocking) fastness measures colour transferred by friction — the white marks on a dark trouser seat, or dye on a white handbag strap. In ISO 105-X12 a standard white cotton rubbing cloth is mounted on a finger of defined diameter and load and rubbed back and forth over the specimen for a set number of cycles in a crockmeter. Two tests are made: with a **dry** rubbing cloth and with a **wet** rubbing cloth (wetted to a defined pick-up). Only staining is graded, with the A03 scale. Wet rubbing is almost always worse — water swells the fibre, lubricates transfer and carries loose pigment — so specifications set a lower minimum for wet (commonly 2–3) than for dry (commonly 4). Deep shades in pigment prints and indigo denim are notorious: denim is expected to crock, and its care labelling warns accordingly; a deep black dress fabric grading wet-rub 2 against a 3 minimum is a reject. Surface colour (pigment, coating, heavily surface-dyed yarn) rubs off far more readily than dye penetrated into the fibre.

TEST	STANDARD	WHAT IS GRADED	SCALE
Light	ISO 105-B02	Colour change vs blue-wool references	1–8 (8 best)
Washing	ISO 105-C06	Colour change + staining of each multifibre band	Grey scales 5–1
Rubbing	ISO 105-X12	Staining of standard cloth, dry and wet	Grey scale A03, 5–1
Colour change (general)	ISO 105-A02	Contrast vs unexposed original	5 (none) to 1 (severe)
Staining (general)	ISO 105-A03	Contrast vs clean white	5 (none) to 1 (severe)

6. Estimating chemical and physical damage

Processing — bleaching, mercerising, resin finishing, faulty storage — can degrade cellulose without any visible change, until the fabric fails in wear or dyeing becomes uneven. Damage estimation detects and roughly quantifies that hidden degradation. The methods below are concept-level laboratory indicators, used comparatively: the same fabric before and after a process, or a suspect lot against a sound control.

INDICATOR	PRINCIPLE	EFFECT OF DAMAGE
Cuprammonium fluidity	Cellulose is dissolved in cuprammonium hydroxide; the fluidity (reciprocal of viscosity) of the solution reflects chain length — shorter chains flow more easily	Fluidity **rises** as chemical damage shortens the cellulose chains; a sound bleached cotton has low fluidity, an over-bleached one markedly higher
Copper number	Degraded cellulose contains more reducing end-groups and oxidised groups, which reduce a standard copper reagent	Copper number **rises** with oxidative and acid damage
Alkali solubility	Degraded, short-chain cellulose dissolves more readily in hot dilute alkali than sound cellulose	Solubility % **rises** with damage
Tensile / tear strength loss	Direct mechanical test before and after processing	Strength **falls**; a loss beyond the mill allowance (often ~10–15 % for a finish) signals over-treatment — but strength alone misses early chemical damage that fluidity already shows
Methylene blue absorption	Oxidised cellulose (oxycellulose) carries acidic groups that absorb the basic dye strongly	Absorption **rises** with oxidative damage; useful for localising tendered patches

Physical damage is estimated differently: abrasion and cutting are seen microscopically as flattened, fibrillated or severed fibres, and heat damage shows as discoloration and brittleness. The diagnostic habit to learn is to pair a chemical indicator with a strength test: fluidity tells *how much* chain degradation has occurred, strength tells *whether it already matters mechanically*, and microscopy tells *where*.

7. Determination of size

7.1 Why size is measured

Warp yarns are coated with size — traditionally starch, nowadays often PVA, acrylic or CMC, alone or blended — to survive weaving. Before dyeing and finishing, size must be removed (desizing); residual size causes resist marks, uneven dyeing and a harsh handle. The size content of a grey fabric, and the completeness of its removal, are therefore routine determinations.

7.2 Identification

The classic spot test uses **iodine**: a drop of iodine solution on the fabric turns **blue (blue-black)** in the presence of **starch** — the familiar starch–iodine reaction. No blue colour means the size is not starch (or starch has already been removed), and further tests distinguish PVA and other synthetics. Identification matters because desizing chemistry must match the size: starch is removed enzymatically or by acid/oxidative treatment; PVA requires hot washing; using the wrong route wastes a process and leaves size behind.

7.3 Desizing and the weighing method

The quantitative method is gravimetric. A specimen is dried and weighed (sized dry mass), then desized completely — for starch, treated with a desizing enzyme bath under its optimum conditions (or dilute acid in the classical method), washed thoroughly until the iodine test on the fabric is negative — dried again and reweighed (desized dry mass). The loss in mass is the size removed. Drying to constant mass at each stage is essential: a 1 % moisture error is larger than many size specifications.

$$\text{Size \% (on desized weight)} = (\text{sized mass} - \text{desized mass}) / \text{desized mass} \times 100$$

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WORKED EXAMPLE — SIZE CONTENT, BOTH CONVENTIONS

Grey fabric specimen, dry mass sized = 50.00 g. After complete enzymatic desizing, washing and drying, dry mass = 44.20 g.
Size removed = 50.00 − 44.20 = **5.80 g**.
On desized weight: 5.80 / 44.20 × 100 = **13.1 %**. On sized weight: 5.80 / 50.00 × 100 = **11.6 %**.
Both numbers describe the same fabric. The desized-weight convention (size add-on relative to the yarn it was applied to) is the weaver’s figure; the sized-weight convention is the fraction of the grey cloth that is size. A report that states only “size = 12 %” without its basis invites dispute — always state both basis and method, and confirm completeness with a negative iodine test before the second weighing.

Two corrections keep the weighing method honest. First, desizing and washing also remove natural impurities and spinning oils, so the mass loss slightly overstates true size — for precise work a control of known unsized yarn is carried through the same treatment. Second, incomplete desizing understates

size: the iodine spot test (for starch) at several places on the specimen is the completeness check, not a single test at the edge.

Glossary

Colour fastness — resistance to colour change and staining under an agency. **Grey scale** — paired-patch scale grading change (A02) or staining (A03) from 5 to 1. **Blue-wool scale** — light-fastness references graded 1–8. **Multifibre fabric** — adjacent strip of different fibre bands used to detect staining in one test. **Crocking** — colour transfer by rubbing. **Fluidity** — flow measure of a cellulose solution; rises with chain degradation. **Copper number** — reducing-power index of degraded cellulose. **Size** — protective coating applied to warp yarn for weaving. **Desizing** — removal of size before wet processing.

Key points

• Every fastness test grades two things: colour change (A02) and staining (A03), 5 = none, 1 = severe.
• Light fastness uses the blue-wool scale 1–8 (ISO 105-B02); each step is roughly double the life.
• Wash fastness (ISO 105-C06) uses a multifibre strip — report the worst staining band; rub fastness (ISO 105-X12) is graded dry and wet, wet always worse.
• Hidden damage: fluidity, copper number and alkali solubility all rise as cellulose degrades; pair them with a strength test.
• Size: iodine turns blue with starch; the weighing method must state its basis — 13.1 % on desized weight is 11.6 % on sized weight for the same fabric.

Practice

1. A specimen dries at 62.50 g sized and 55.80 g after complete desizing. Calculate the size removed and the size % on both reporting bases.
2. A navy fabric grades: wash change 4, worst multifibre staining 3 (nylon band), dry rub 4, wet rub 2–3, light 4. The specification requires change ≥ 4, staining ≥ 3–4, dry rub ≥ 4, wet rub ≥ 3, light ≥ 4. Decide pass or fail and list every failing item.
3. Why is wet rubbing fastness almost always lower than dry rubbing fastness for the same fabric?
4. An over-bleached cotton shows fluidity well above the sound control but only a small strength loss. Explain why both results can be true and which one predicts future tendering complaints.

Answer key

1. Size removed = 62.50 − 55.80 = **6.70 g**. On desized weight: 6.70 / 55.80 × 100 = **12.0 %**. On sized weight: 6.70 / 62.50 × 100 = **10.7 %**.

2. **Fail.** Wash change 4 meets ≥ 4 (pass). Staining 3 is below the ≥ 3–4 minimum (fail). Dry rub 4 passes. Wet rub 2–3 is below ≥ 3 (fail). Light 4 passes. Two failures — staining and wet rub — either alone justifies rejection or a negotiated concession with price/usage limits.
3. Water swells the fibre and softens the surface, dissolves or disperses loose surface dye and pigment, and acts as a transfer medium carrying colour onto the rubbing cloth. The same loose colour that stays put when dry is readily picked up when wet, so the wet grade is typically one to two grades lower.
4. Fluidity measures average chain length in solution — it detects chain scission as soon as it occurs, throughout the fibre. Fabric strength is carried by the surviving load-bearing structure and falls noticeably only when scission is extensive or localised at critical places. Early chemical damage therefore registers in fluidity first; the fabric will tender later in wear and laundering as the shortened chains fail progressively. The fluidity result is the early warning and the better predictor of tendering complaints.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib